

Advanced Stats

Statistics: A branch of applied mathematics that involves the collection, description, analysis and inference of conclusions from qualitative and quantitative data.

Experiment: Process that generates information.

Deterministic Exp: when you get same outcome every time you perform an exp.

① **Random Exp:** when you get different outcome every time you perform.

② **Sample Space:** Set of all possible outcomes written in the form of ordered pairs (order matters).

③ **Event:** Subset of whole sample space.

④ **Probability:** It quantifies chance of occurrence from random experiment.

Axioms:

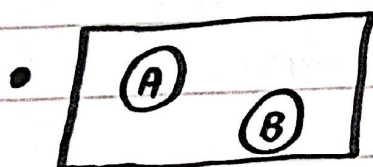
① $0 \leq P(A) \leq 1$

② $P(S) = 1$ (outcome should be from SS)

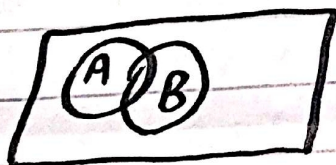
③ if $A_1, A_2 \dots A_N$ are disjoint events

④ $P(A_1 \cup A_2 \cup A_3 \dots \cup A_N) = P(A_1) + P(A_2) \dots P(A_N)$

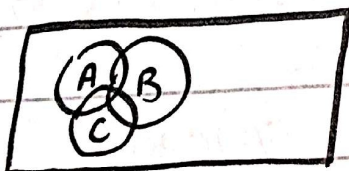
if you are dealing with an experiment in which everything has equally likely outcome then, Probability of an event can be number of points in set A (events) and number of points in sample space (S). $n(A) / n(S)$



$$P(A \cup B) = P(A) + P(B)$$



$$P(A) + P(B) - P(A \cap B)$$



$$P(\underbrace{A \cup B}_{\text{set 1}} \cup \underbrace{C}_{\text{set 2}})$$

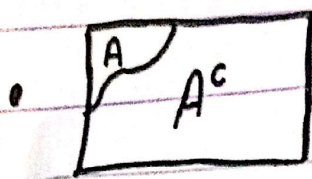
$$P[A \cup (B \cup C)]$$

$$P(A) + P(B \cup C) - P[A \cap (B \cup C)]$$

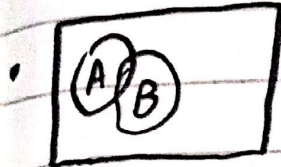
$$P(A) + P(B) + P(C) - P(B \cap C) - P[(A \cap B) \cup (A \cap C)]$$

$$P(A) + P(B) + P(C) - P(B \cap C) - [P(A \cap B) + P(A \cap C) - P(A \cap B \cap C)]$$

$$P(A) + P(B) + P(C) - P(B \cap C) - P(A \cap B) - P(A \cap C) + P(A \cap B \cap C)$$



$$P(A) = 1 - P(A^c)$$

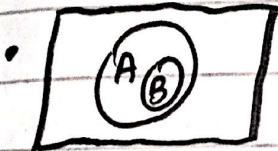


A given find B

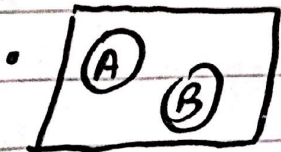
$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

Conditional Prob
tells Joint Prob.

$$P(A \cap B) = P(B/A) P(A)$$



$$P(A/B) = 1$$



$$P(B/A) = 0$$

Question: Conditional Prob.

Data: 0.10 fail length test
0.05 fail texture test
0.008 fail both.

let

L: length defective, T: texture defective.

$$P(L) = 0.1$$

$$P(T) = 0.05$$

$$P(T \cap L) = 0.008$$

$$P\left(\frac{T}{L}\right) = \frac{0.008}{0.1} = 0.08.$$

$\Rightarrow$ if occurrence of A does not depend occurrence of B.

$$P(B|A) = P(B) \quad \text{--- (1)}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \quad \text{--- (2)}$$

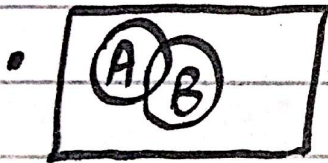
$$\frac{P(A \cap B)}{P(A)} = P(B)$$

$$P(A \cap B) = P(A) \cdot P(B) \quad (\text{Independent Events})$$

This formula is used to check whether the events are independent or not.

General formula:

$$P(A_1 \cap A_2 \dots \cap A_n) = P(A_1) \cdot P(A_2) \dots P(A_n).$$

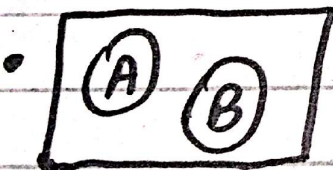


$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cap B) = P(A|B) P(B)$$

or

$$P(B|A) P(A)$$

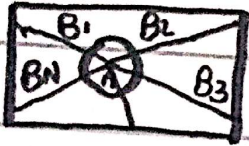


$$P(A \cup B) = P(A) + P(B)$$

$$P(A \cap B) = 0.$$

Note: All mutually exclusive events are not independent.

BAYE'S THEOREM:



$$A = (A \cap B_1) \cup (A \cap B_2) \cup (A \cap B_N)$$

$$P(A) = P[(A \cap B_1) \cup (A \cap B_2) \cup \dots (A \cap B_N)]$$

$$= P(A \cap B_1) + P(A \cap B_2) + P(A \cap B_N)$$

$$= P\left(\frac{A}{B_1}\right) P(B_1) + P\left(\frac{A}{B_2}\right) P(B_2) + \dots P\left(\frac{A}{B_N}\right) P(B_N)$$

$$= \sum_{i=1}^N P\left(\frac{A}{B_i}\right) P(B_i) \text{ (Total Prob law)}$$

2.41 Example:

$$P(B_1) = 0.30$$

B - Machine No. Product

$$P(B_2) = 0.45$$

$$P(B_3) = 0.25$$

Defective products = A

$$P(A) = ?$$

$$P\left(\frac{A}{B_1}\right) = 0.02$$

$$P\left(\frac{A}{B_2}\right) = 0.03$$

$$P\left(\frac{A}{B_3}\right) = 0.02$$

$$\text{Total } P(A) = P\left(\frac{A}{B_1}\right) P(B_1) + P\left(\frac{A}{B_2}\right) P(B_2) + P\left(\frac{A}{B_3}\right) P(B_3)$$
$$= (0.02)(0.30) + (0.03)(0.45) + (0.02)(0.25)$$

$$P(A) = 0.0245$$

Defective

By Baye's Theorem we have to find

$$P(B_3|A)$$

$$= \frac{P(A \cap B_3)}{P(A)} = \frac{P(A|B_3) P(B_3)}{P(A)}$$

$$= \frac{(0.02)(0.25)}{0.0245} = \frac{10}{49}$$

Practice questions: 2.123, 2.106, 2.120,
2.96, 2.98.
